



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

FURTHER MATHEMATICS

9187/2

PAPER 2

NOVEMBER 2012 SESSION

3 hours

Additional materials:

Answer paper

Graph paper

List of Formulae

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81 \text{ ms}^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Within each section of the paper, questions are printed in the order of their mark allocations and candidates are advised, within each section, to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

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[Turn over]

Section (a) : Mechanics

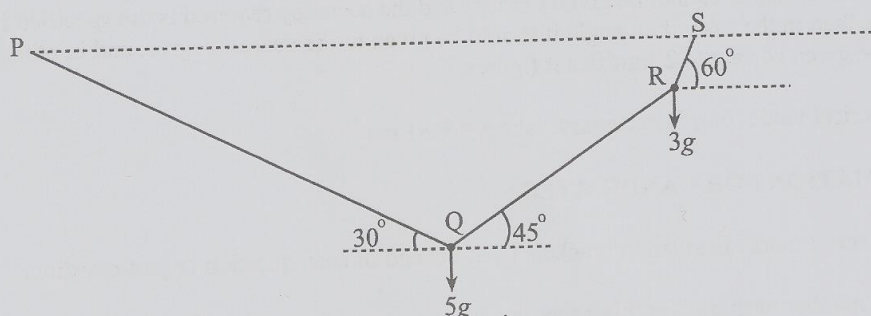
- 1 A student pushes a ball bearing of mass $2g$ on a smooth horizontal glass table. Experimental evidence suggests that the resistive force is $0.008v^3$ where v is the velocity of the ball bearing.
- (i) Find an expression for the velocity, v , of the ball bearing in terms of the displacement s , given that it starts from the origin with velocity 4 cm s^{-1} . [5]
- (ii) Find the distance in which the speed of the ball bearing falls from 2 cm s^{-1} to 0.75 cm s^{-1} . [2]

- 2 At a Harare Showground stand, a game is won if one hits a bead threaded on a vertical smooth circular ring so that it completes a circle on the ring.

Given that the mass of the bead is 0.02 kg and the ring has a radius of 0.4 m , one end of the elastic string has the bead fixed to it and the bead is free to rotate round the ring. The other end of the elastic string is fixed to the lowest point P at the ring. The elastic string has a natural length of 0.4 m and modulus of elasticity 0.6 N .

Find the minimum horizontal velocity for which the bead must be projected with from P , in order to win the game. [7]

3



A light inextensible string PQRS has two particles of mass 5 kg and 3 kg attached at Q and R . The ends of the strings are attached to points P and S at the same horizontal level (see diagram above).

In the equilibrium position PQ , QR and RS are inclined at angles of 30° , 45° and 60° respectively to the horizontal.

Find the tensions in the string. [7]

- 4 An elastic string has natural length 2 m and modulus of elasticity 1.96 N. The ends of the string are fixed at two points P and Q which are 3 m apart on a smooth horizontal table.

A particle of mass 0.02 kg is attached to the mid point of the string. The particle is pulled towards Q and is released at a distance 2 m from P.

- (i) Show that the equation of the subsequent simple harmonic motion is

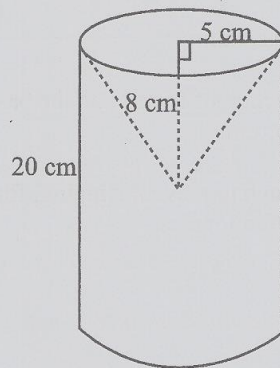
$$\frac{d^2x}{dt^2} = -196x$$

where x is the displacement from the mid point of the string. [3]

- (ii) Find the

1. period of the particle,
2. amplitude of the particle,
3. speed of the particle at 1.5 m from P. [5]

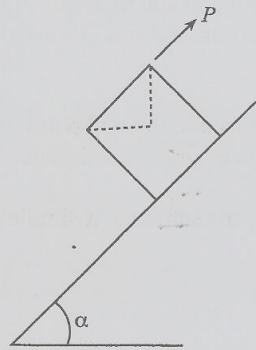
- 5 (a)



A solid of height 20 cm and radius 5 cm has a right cone of height 8 cm and base radius of 5 cm cut from it symmetrically (see diagram above).

Find the centre of gravity of the remaining solid. [4]

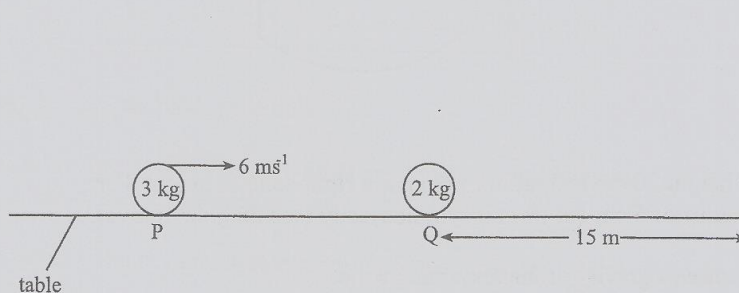
(b)



The remaining solid from 5 (a) is now placed on a rough inclined plane with an angle of inclination α to the horizontal with the uncut side on the plane. A force of magnitude P is applied to the top of the solid upwards and parallel to the plane (see diagram).

- (i) Giving your answers in terms of W (the weight of the solid), α and P , find the
1. force of friction,
 2. normal reaction,
 3. point of action of friction on the base of the solid if the system is in equilibrium.
- (ii) If the plane is rough to prevent slipping, find the maximum force P such that the solid topples. [10]

6



Two particles P and Q of masses 3 kg and 2 kg respectively lie on a smooth horizontal table. P is projected with a speed of 6 ms^{-1} directly towards Q which is at rest (see diagram).

- (a) If the coefficient of restitution is $\frac{2}{3}$, find the speed of $\dot{Q}$ immediately after impact. [4]

- (b) After travelling a distance of 15 m, Q strikes a vertical wall directly and rebounds. The coefficient of restitution between the wall and Q is also $\frac{2}{3}$.

Find how far Q travels before it next collides with P again. [6]

- (c) Calculate the loss in the kinetic energy at the first impact. [2]

- (d) Find the impulse between the two particles on their second impact. [5]

Section (b): Statistics

- 7 Three machines in a production line fill up bottles of a beverage. Amounts filled by the three machines are normally distributed with parameters given below

machine	mean	standard deviation
P	345	5
Q	350	4
R	340	2.5

A quality inspector samples 10 bottles from machine P, 5 from machine Q and 8 from machine R.

Calculate the distribution of the combined sampled quantity. [3]

- 8 A sample of 13 results for a typical A level Mathematics class were recorded as

15, 63, 45, 10, 36, 44, 72, 65, 51, 60, 49, 38, 40

Find the unbiased estimates of the population mean and variance. [3]

- 9 A cook at a restaurant is concerned that eggs delivered to him as Standard by a local farm are in fact small. He measures the weight of a random dozen of eggs, and the results in grammes are

65, 58, 70, 85, 72, 55, 87, 68, 56, 48, 45, 57.

Given that the weight of standard eggs is distributed normally with a mean 70 grammes, use 5% level of significance to prove this or otherwise. [6]

- 10 (a) Two typists A and B can make errors on a page when typing. The number of errors made by the typists are Poisson distributed with means $\frac{3}{4}$ and $\frac{1}{2}$ respectively.
Find the probability that more than three errors are made by the two typists per page. [3]

- (b) A rare cattle disease affects one in every two thousand cattle.
Using a suitable approximation, find the probability that less than two in a farmer's herd of 80 cows are affected by the disease. [3]

- 11 A company is introducing a new energy saving bulb on the market. It claims that the lifespan, x measured in years of the bulb, is given by the distribution

$$f(x) = 2e^{-2x}, \quad x > 0.$$

A study of 5 400 such bulbs yielded the following results.

lifetime	$0 \leq x < 1$	$1 \leq x < 2$	$2 \leq x < 3$	$3 \leq x < 4$	$x \geq 4$
frequency	4650	635	103	10	2

Test the company's claim at 1 % level of significance. [13]

- 12 The tensile strength of a material is measured for rods of varying diameter. The results are shown in the table below:

Diameter, D	7.3	8.8	10.3	11.4	17.0	16.4	19.1	12.5	9.4	8.8
Stress, T	10.7	11.3	9.4	12.5	15.3	16.4	18.0	10.9	7.4	8.3

- (a) Make a scatter plot of the data. [4]
- (b) Find the regression line of **T** on **D** and then sketch the regression line on the same axes as the scatter plot. [5]
- (c) Calculate the product moment correlation coefficient. Comment on your result. [3]
- (d) A material scientist wants to make a rod with a diameter of 11, find the tensile stress you would recommend. [2]

- 13 (a) A discrete random variable R has the probability density function

$$f(r) = \frac{r}{15}, \quad r = 1; 2; 3; 4; 5.$$

- (i) Find the probability generating function of R .
 (ii) Use your result to find the mean and variance of R .

[7]

- (b) If X is geometrically distributed with probability density function

$$P(X = r) = q^{r-1}p \quad r = 0, 1, 2, \dots$$

where $q = 1 - p$ and $0 < p < 1$, show that the probability generating function $G(t)$ is given by

$$G(t) = \frac{pt}{1 - qt}. \quad [4]$$

- (c) Use the result above to find the mean and variance of the geometric distribution with parameter $p = \frac{2}{5}$.

[4]